

NICOLAS VEGA

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SUMMARY

Full-stack engineer building LLM agent infrastructure: production browser agents with safety enforced at the tool layer, multi-agent orchestration, and retrieval pipelines for agent context. Ships 0→1 products in regulated environments (HIPAA, insurance, EDI). UC Berkeley CS '25.

PROFESSIONAL EXPERIENCE

Ambra911 (HIPAA revenue-cycle platform for EMS providers)

July 2026 – August 2026

Founding Engineer

San Francisco, CA

- **Shipped 0→1 a payer-agnostic claims-processing product** spanning claim statusing, denials management, and payer follow-up. Go/PostgreSQL pipeline issuing X12 EDI 276 requests and parsing 277 responses through the Stedi clearinghouse, with multi-source ingestion (837 uploads, billing sync, SFTP), canonical status mapping, and per-payer recheck scheduling. 65 merged PRs and 46 database migrations in 4 weeks.
- **Replaced a third-party vendor with a production Playwright + Claude browser agent** (~4k LOC TypeScript, sole author, 11 days) that files appeal documents on a payer portal: deterministic Playwright flow with an LLM fallback that activates only when fragile steps fail, driving thin tools (read_page, click, type, upload_files).
- **Designed tool-layer safety guarantees for an irreversible action:** final submission gated four independent ways, with the submit tool never registered with the model unless every gate passes, enforcing never-submit at the tool layer rather than in the prompt. Shipped a signed live e2e harness that exercises the full flow and stops before submission.
- **Built an AI voice agent** that autonomously phones insurance payers when electronic status is unsupported (Retell, OpenAI, GCP Cloud Tasks/Scheduler): business-hours gating, AI-disclosure compliance, idempotent retry, and an LLM transcript extractor normalizing rep speech into the same structured status vocabulary as the EDI path.
- **Built autonomous agent loops for error remediation:** production failures triaged and turned into fix PRs without human authoring, closing the loop between runtime errors and code changes.
- **Built the claims and denials operations frontend** (Next.js 15, React 19, TypeScript): cross-practice control plane, automated check queue, EDI 837/835 upload with async remittance-parse polling, and configurable denials worklists, backed by ~1.7k LOC of Playwright e2e tests.
- **Hardened platform security:** fail-closed webhook authentication with replay-window and SSRF guards, closed multiple PHI-leak vectors, PHI-masked agent artifacts to a short-TTL CMEK bucket, and a weekly automated OWASP ZAP DAST pipeline routed to Slack and archived as SOC 2 evidence, alongside Trivy/Semgrep CI gates.

Agentic IDE Inc. | aideapp.dev

March 2026 – May 2026

Founder & Lead Developer

San Francisco, CA

- **Founded Agentic IDE to attack the technical debt AI coding agents introduce:** shipped an MCP-server harness with pre-plan hooks exposing a plan_mode_skill_audit tool that autonomously maintains a team's skill library, proposing new skills and upgrading stale ones as the codebase evolves.
- **Designed skills as embedded, schema'd objects** supporting graph search over relational data, with recursive agent optimization (RAO) for retrieval and background skill evolution via a lightweight GEPA protocol on Hermes.
- **Built Aether, an autonomous agent brain** chaining evolved skill profiles into composable agent jobs and performing access-scoped RAG over memories and skills, enabling long-lived agent loops and graphs that continuously develop and maintain an application.
- **Closed an enterprise pilot with Arena AI** (\$1.7B, 50+ engineers) as a solo founder; the deal lapsed when SOC 2 certification proved unfundable pre-seed.
- **Shipped to production in 21 days and scaled to 250+ concurrent users** on Azure Container Apps backend, Vercel frontend, Stripe billing, and published npm packages. Backed by Microsoft for Startups within the first week and Founders Inc (Founders Offseason 2026).

Corgi (insurtech)

January 2026 – February 2026

Full Stack Engineer (Contract)

San Francisco, CA

- **Owned the Shopify integration and B2C pipeline 0→1:** shipped the customer-facing Shopify app in 2 days, then absorbed a backend refactor preserving the monolith through two weeks of regression testing and continuous shipping to unblock teammates, to production deploy and hand-off. CTO attributed \$10M in immediate revenue and \$100M+ in potential across airline, shipping, and consumer-product verticals.
- **Owned the Bland AI call-ingestion pipeline:** JSON webhook ingestion, a Python processing and transformation layer, and .NET API translation for subsidiary systems, enabling automated customer intake and real-time data flow across heterogeneous stacks.
- **Owned growth and analytics pipelines** and built in-house lead-generation algorithms.
- **Contributed AI workflows and ingestion layers to the underwriting system:** built a reusable AI tooling framework for rapid integration across products, resolved 12 critical issues in the production multi-agent intake system, and implemented an automated trucking insurance application with OCR document processing.
- **Handled claims processing for the shipping and trucking insurance verticals:** built an 82%-accuracy classifier for approve/partial/deny decisions and an XGBoost regression model with ChromaDB vector embeddings achieving <15% RMSE on payout calculation, automating the underwriting workflow end to end.

YNLD Trust

November 2025 – January 2026

AI Engineer (Contract)

San Francisco, CA

- **Built an AI trading-decision workflow** ingesting market data from external APIs, routing it through the firm's legacy internal evaluation model, and producing buy/sell/hold judgments, gated by evals and a multi-stage governance chain scoring risk factors before any recommendation is issued. 10% average return against the evaluation suite.
- **Built a transcript-synthesis workflow** turning YouTube transcripts into structured summaries, deployed standalone and as a feed into the agentic ingestion pipeline; incorporated news sentiment as an additional model signal.

- **Engineered a production Chrome extension** (Manifest V3) processing 100+ playlists asynchronously via service workers and offscreen documents to work around browser timeout limits.

Wogo Life Inc. | thewogo.com

September 2025 – January 2026

Software Engineer (Contract)

Remote

- **Solo-architected and launched a cross-platform social platform** across web, iOS, and Android (React, React Native, TypeScript, Swift, AWS serverless: Lambda, DynamoDB, S3, CloudFront), with 99.9% uptime at under \$100/month in infrastructure cost, shipped in 3 months as sole developer for a 10% equity stake.
- **Built the video pipeline and API infrastructure:** serverless video processing with FFmpeg on Lambda (95% upload-time reduction), 40+ REST APIs with JWT auth via AWS Cognito, WebSocket messaging, Google Maps integration, and CI/CD.
- **Architected 20+ services** spanning login, feed, video upload, geolocation, live chat messaging, user profiles, network reliability, and authorization.

Farmers Insurance

June 2023 – August 2023

Software Engineer Intern

Remote

- **Saved \$504K annually** by designing an API gateway workaround for Salesforce live chat, avoiding a \$42K/month license while enabling data flow to an Oracle database, deployed nationally across all US subsidiaries.
- **Architected a Salesforce-to-Oracle data pipeline** with a GCP API gateway and ETL, scaling from hundreds to 10,000+ concurrent users depending on payload size, and built compliance dashboards working across product, IT, and security teams.

PROJECTS

Personal-Harness / BetterAI

github.com/NicoV7/Personal-Harness

- Sole author, 108 commits. Personal coding-agent harness: plan-scope gating, a skill-corpus rules engine, lifecycle hooks, containerized multi-arch deploys, and eval tooling. A retrieval pipeline (hybrid keyword search plus bge-small-en embeddings, cached in Redis Vector Library) fires when the harness finishes planning, matches the implementation plan against an indexed skill corpus, and injects the matched rules into implementation, including for spawned subagents. Forked into my employer's engineering org and adopted with private skill-corpus sync, August 2026. Public skill corpus at github.com/NicoV7/betterai-skills.

Yojimbo

yojimbo.site

- Autonomous AI investment-research agent built on Aether, covering market analysis, SEC filing synthesis, and risk assessment, with a public agent performance leaderboard. Uses profit as the goal metric for long-running autonomous agent tasks.

Debate RPG (Berkeley AI Hackathon 2026)

github.com/NicoV7/BerkeleyAIHackathon2026

- Local-first creature-collector RPG where the monsters are autonomous debate agents: players collect agents, give them gambit-style behavior rules, train their prompt genomes, and send them into turn-based argument battles against 1–3 enemy LLM personas. Vite/React/Phaser 3 client, FastAPI backend, Postgres + pgvector for agent memory and hybrid RAG, Redis for live encounter cache.
- **Built a latency-first model gateway** routing battle actors and judges across hosted providers (Groq, Cerebras, Gemini, OpenRouter) with local Ollama fallback, selecting providers from a benchmarked Pareto frontier refreshed by an in-process benchmarking script.
- **Built a self-play prompt-genome training loop** evolving party and enemy agents against each other, scoring latency first and accepting only mutations that clear quality and reliability floors.

LearnGraph (2nd place, GBrain Hackathon, hosted by Y Combinator)

- AI-powered learning app turning any topic into an interactive knowledge graph, built 0→1 in 8 hours on a pipeline of 11 distributed agent tasks: web research, topic decomposition, orchestration, parallel lesson writing, and quiz generation.

EDUCATION

University of California, Berkeley, B.A. Computer Science

August 2022 – May 2025

GPA: 3.31

City College of San Francisco

June 2019 – May 2022

A.S. Computer Science · A.S. Mathematics · A.S. Liberal Arts and Sciences · GPA: 3.94

SKILLS

Languages: Go, TypeScript, JavaScript, Python, SQL, Java, C/C++, C#, .NET, Swift, HTML/CSS

Agents & AI: Anthropic SDK, OpenAI, Retell, Bland AI, MCP, tool-use design, multi-agent orchestration, RAG (hybrid, graph, agentic), pgvector, ChromaDB, Redis Vector Library, scikit-learn, XGBoost

Backend: Gin, GORM, Fastify, FastAPI, Django, Node.js, Celery, REST, GraphQL, WebSockets, PostgreSQL, Redis, MongoDB, DynamoDB, RabbitMQ, X12 EDI

Frontend: React 19, Next.js 15, React Native, TypeScript, Tailwind, Radix/shadcn, Zustand

Cloud & Infra: GCP (Cloud Run, Tasks, Scheduler, GCS), AWS (Lambda, Cognito, S3, CloudFront), Azure Container Apps, Docker, GitHub Actions, CI/CD, Vercel, Linux

Testing & Security: Playwright, Vitest, JUnit, TDD, OWASP ZAP, Trivy, Semgrep, HIPAA and SOC 2 compliance